

Sravya Nallagantula

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Profile Summary

Junior AI Engineer with experience building **Generative AI applications**, **LLM-based workflows**, and production-ready **AI APIs** for real time customer and business automation. Skilled in **Python**, **FastAPI**, **SQL**, **Docker**, **AWS**, **Azure**, and retrieval architectures using **FAISS**, **Pinecone**, and **Hugging Face**. Strong foundation in **RAG pipelines**, **AI agents**, **prompt engineering**, **LLM evaluation**, **MLOps**, **CI/CD**, and model monitoring, with a proven ability to deploy, troubleshoot, and optimize AI-powered applications.

Education

Master of Science in Computer and Information Sciences , Concordia University, Mequon, WI	2023 – 2025
Bachelor of Science. , Keshav Memorial Institute of Commerce and Sciences, Hyderabad, India	2017 – 2020

Work Experience

Optum AI Engineer	May 2025 – Present Boston, US
- Reduced 40% of manual support work by automating 100+ customer-style intake calls/week through a production-ready GenAI workflow built using Python, FastAPI, OpenAI, VAPI, and ElevenLabs. - Enhanced contextual response accuracy by 15% by building a LangGraph based AI agent workflow using LLM APIs, prompt engineering, tool calling, and FastAPI integrations, enabling intelligent automation and reliable outputs. - Improved grounding reliability by 25% by building a hybrid SERP plus full-site scraping pipeline using SerpAPI, BeautifulSoup, embeddings, semantic search, and vector databases, enabling stronger RAG pipelines. - Cut application latency by 28% by optimizing asynchronous FastAPI routes, adding Redis caching, and restructuring inference flows across containerized Azure services, enabling faster APIs, performance optimization, and stable deployment. - Achieved p95 latency under 800ms by load testing from 1 to 5000 users using Postman, Python scripts, and CI/CD checks to benchmark throughput, evaluate outputs, troubleshoot issues, and stabilize AI applications.	
Accenture Data Engineer	Aug 2022 – Aug 2023 Hyderabad, India
- Reduced manual review effort by 30% by building automated data quality checks across production workflows using SQL, Python, and PySpark, validating datasets for model development and exception reporting. - Boosted pipeline performance by 6% by applying SQL optimization across Snowflake and database workflows, improving query speed, schema reliability, transformation logic, and downstream AI-ready data processing. - Minimized data quality failures by 25% by preparing, cleaning, transforming, and analyzing multi-source datasets using SQL, Python, and PySpark, owning validation checks and recurring pipeline refreshes. - Resolved 100+ production issues by owning end-to-end Airflow job monitoring, coordinating Agile fixes in Jira, publishing curated outputs to AWS and Snowflake, and adding logging, rollback runbooks, and documentation.	
Accenture Machine Learning Engineer	Apr 2021 – Aug 2022 Hyderabad, India
- Reduced model validation time by 40% by automating end-to-end ML evaluation pipelines using Python, GitHub Actions, MLflow, and regression checks, enabling model monitoring, testing, and release validation. - Improved model accuracy and training consistency by combining TensorFlow, Keras, scikit-learn, NumPy, Pandas, and SQL workflows to build NLP and computer vision pipelines, tune models, and validate features.	

Skills

- Programming and APIs:** Python, SQL, JavaScript, TypeScript, FastAPI, REST APIs, Git, GitHub, Linux
- Generative AI and LLMs:** OpenAI APIs, Hugging Face, LLMs, Prompt Engineering, LLM Evaluation, AI Workflows
- AI Agents and RAG:** LangGraph, LangChain, AI Agents, Tool Calling, RAG, FAISS, Pinecone, Vector Databases
- Machine Learning and Deep Learning:** TensorFlow, PyTorch, Keras, Scikit Learn, NLP, Computer Vision, Model Optimization
- Data Engineering and Databases:** PySpark, Databricks, Spark SQL, ETL Pipelines, Airflow, Snowflake, Pandas, NumPy
- Cloud and Deployment:** AWS, Azure, Docker, Kubernetes, Microservices, Cloud Deployment, Performance Optimization
- MLOps and Production Support:** CI/CD, GitHub Actions, MLflow, Model Monitoring, Testing, Troubleshooting, Documentation

Projects

CreditWise-AI - RAG and LLM Assistant: Built a **FAISS-based GenAI application** using **LangChain**, **Hugging Face**, and **Groq** to deliver document-aware answers, improving retrieval precision by **38%** through semantic reranking and optimized embeddings.

Real Estate Voice Agent - AI Automation Workflow: Built a **production-ready AI agent** using **VAPI**, **LangGraph**, and **FastAPI**, orchestrating workflows to capture user intent, extract preferences, evaluate responses, and query **1K+ listings**.